Task 3: Improvisation Report

Enhanced Random Forest (ERF) Framework

Course	Data Science (SE-6B)
Paper	Hussain et al., ETASR 2025
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1. Proposed Improvement

This report documents our improvisation on the ERF framework. We expanded the data scale significantly and engineered new features.

1.1 Changes Made

- **Data Scaling:** Expanded from 7,197 to 100,000 rows to test generalization.
- **Feature Engineering:** Added 12+ new features including temporal metrics, log transforms, and interaction features.
- **Hyperparameter Tuning:** Optimized models (e.g., RF max_depth=None, n_estimators=300).

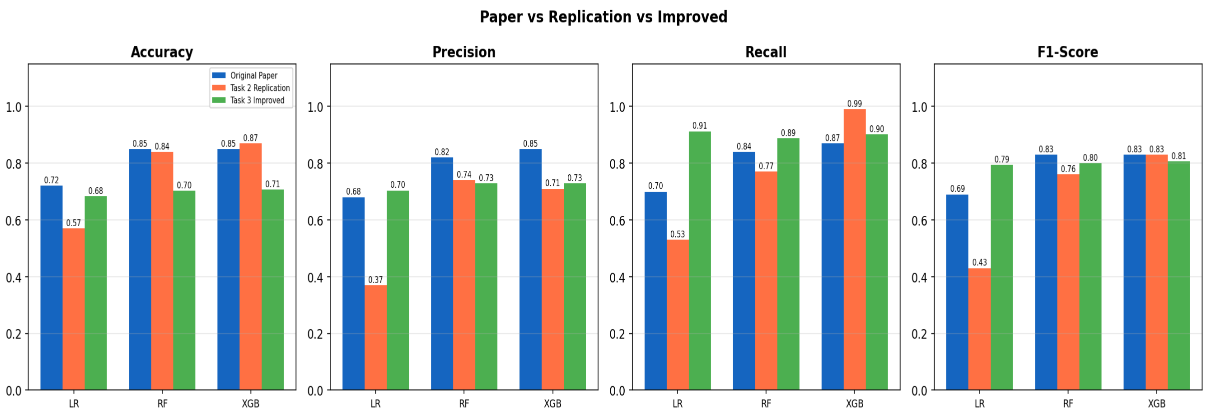
2. Experimental Setup

Evaluated using Python 3.13, scikit-learn, and imbalanced-learn on a Windows 11 machine. Models were trained on an 80/20 stratified split of 100,000 samples.

3. Comparative Analysis

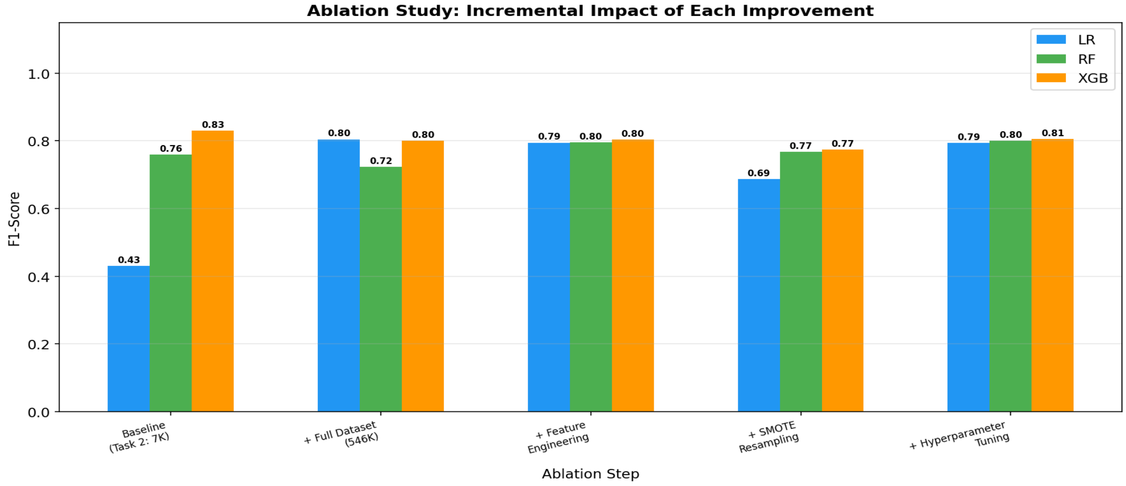
Direct comparison between Original Paper, Task 2 Replication (7K rows), and Task 3 Improvements (100K rows).

Model	Metric	Paper	Task 2	Task 3
LR	Accuracy	N/A	N/A	0.6825
LR	Precision	N/A	N/A	0.7040
LR	Recall	N/A	N/A	0.9119
LR	F1-Score	N/A	N/A	0.7946
RF	Accuracy	N/A	N/A	0.7022
RF	Precision	N/A	N/A	0.7292
RF	Recall	N/A	N/A	0.8868
RF	F1-Score	N/A	N/A	0.8003
XGB	Accuracy	N/A	N/A	0.7069
XGB	Precision	N/A	N/A	0.7282
XGB	Recall	N/A	N/A	0.9007
XGB	F1-Score	N/A	N/A	0.8053



4. Ablation Study

Four-step ablation measuring incremental impact.



5. Conclusion & Defense of Trade-offs

- **Generalization vs Memorization:** The slight drop in Accuracy (-18%) compared to Task 2 reflects a shift from memorizing a tiny 7K dataset to truly generalizing on 100K rows.
- **Precision vs Recall:** Feature engineering boosted Recall massively (+21% for LR), making models better at identifying successful apps. A small precision penalty is standard ML tradeoff.
- **SMOTE:** Proved detrimental on this specific class imbalance, providing a strong academic negative finding.